more clearly the importance of ion clustering processes in the formation of the Venus ultraviolet haze layer.

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Anomalous Palaeolatitudes in Cretaceous Granitic Rocks

IN a recent study of the distribution of fusulinaceans in western Canada, Monger and Ross[1] interpreted their observations in terms of three plate-tectonic models, two of which invoke or might involve large northward translations of portions of the western Cordillera relative to cratonic North America. In this communication we describe anomalously low palaeomagnetic inclinations in radiometrically dated batholithic rocks exposed in the Cascade Range of central Washington. If interpreted in accordance with the usual axial dipole hypothesis of palaeomagnetism[2], our results suggest that these rocks originated roughly 25° further from the Cretaceous pole than their present location relative to the North American craton suggests. A simple application of plate tectonics based on existing models[1,3,4] can tentatively account for our results. Alternative explanations seem unlikely.

Fig. 1 shows the sampling area, near Stevens Pass in the Cascade Range of central Washington. Of greatest interest are the granitic rocks that underlie most of the area; these represent the southern end of an almost unbroken belt of similar plutonic rocks that extends north-north-west along the length of British Columbia into the Yukon and Alaska. Geological mapping and several radiometric ages show that two distinct generations of plutonics rocks are represented in the area of Fig. 1, Tertiary

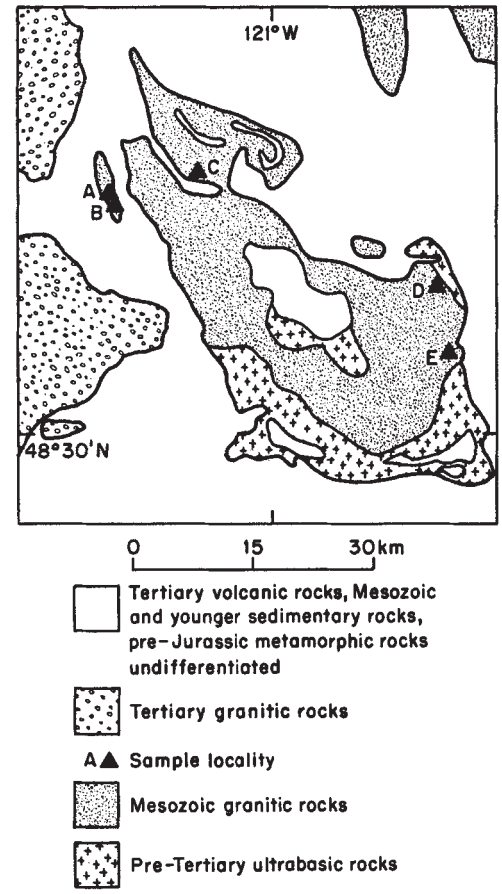


Fig. 1 Location map, modified from the Geological Map of Washington. Sampling localities are described in Table 1.

(ages about 15–50 m.y.) and Late Cretaceous (about 85 m.y.)[5]. Rock types represented typically are massive, medium-grained hypidiomorphic granular plutonic rocks ranging in composition from quartz monzonite to diorite. Mafic and ultramafic rocks are also present but have not been sampled for palaeomagnetic studies. Tectonic complications might be difficult to detect in these massive granitic rocks but seem to be minimal. Exposures are fresh because of recent glaciation and an abundance of artificial exposures. There are two recent descriptions of the geology of the western Stevens Pass area[6,7] and Misch[8] summarizes the complicated regional geology of the northern Cascade Range.

Samples from five sites of Cretaceous age have been studied so far (location in Fig. 1). Samples were drilled and oriented *in situ* using a gasoline powered drill and magnetic compass. Measurements of remanent magnetization were made on a

Table 1 Directions of Remanent Magnetization and Palaeomagnetic Poles

		NRM					After a.f. magnetic cleaning							
Site	N	D	I	κ	α_{95}	$\tilde{H}$	D	I	κ	α_{95}	λ'	φ'	δm	δp
A	15	358.5°	52.0°	30.8	7.0°	150	353.5°	46.5°	63.3°	4.8°	69N	76E	6.2°	4.0°
B	8	263°	71.5°	5.3	26.5°	500	348°	43°	5.4	265°	65.5N	86.5E	32.6°	20.3°
C	10		Unstable											
D	5	352.5°	54.5°	64.8	9.6°	200	350.5°	51°	57.5	10.2°	72N	86.5E	13.8°	9.3°
E	9	15.5°	52°	53.3	7.1°	200	16.5°	52°	57.6	6.8°	68.5N	17.5E	9.2°	6.1°

N, Samples per site; D and I, declination and inclination; κ, estimate of precision parameter[10]; α_{95}, radius of circle of confidence at 95% probability; $\tilde{H}$, peak value of alternating field (oersted); λ' and φ', latitude and longitude of palaeomagnetic pole; δm and δp, semi-axes of oval of 95% confidence about pole. Site locations: *A*, along logging road, south centre sec. 21, 26° N, 12° E; *B*, road cut on Hiway 2 centre sec. 20, 26° N, 12° E; *C*, south end of big road cut on Hiway 2, sec. 15, 26° N, 13° E; *D*, road cut in Tunwater Canyon, Hiway 2, centre sec. 28, 25° N, 17° E; *E*, road cut on Icycle Creek, centre sec. 27, 24° N, 17° E.

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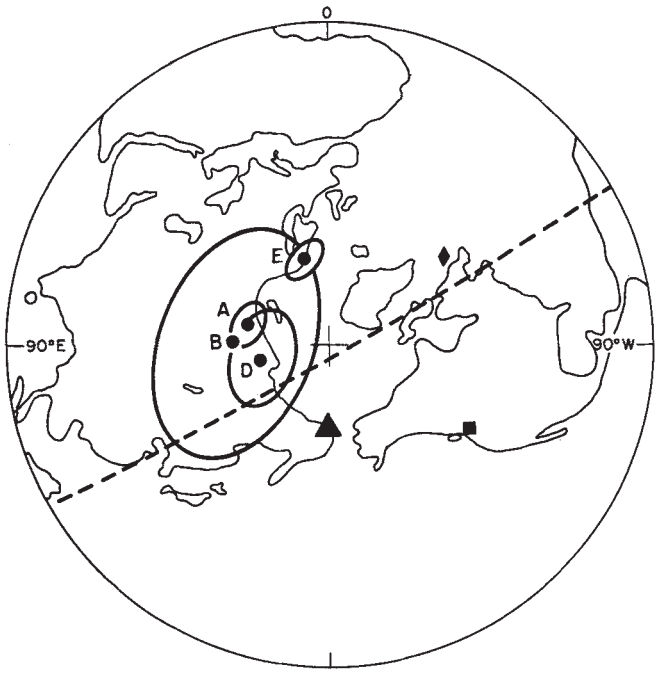


Fig. 2 ●, Polar equal area projection showing poles from Table 1; ▲, mean Cretaceous pole for North America; ◆, pole of rotation of the Pacific relative to North America[22]; ■, sampling locality; - - -, locus of rotation axes capable of transforming the mean North American Cretaceous pole into the mean pole for sites *A*, *D* and *E* (59.5° E, 72.5° N).

Schonstedt SM-1A spinner magnetometer and alternating field demagnetization experiments were performed on a unit of 1,500 oersted peak capacity fitted with a three-axis tumbler similar to the apparatus described by Doell and Cox[9]. All specimens were progressively cleaned in peak fields of up to 200 oersted and selected specimens were further cleaned to 800 oersted. Results of NRM measurements and magnetic cleaning experiments are summarized in Table 1.

Three sites (*A*, *D* and *E*) show high magnetic stability, one site (*B*) is marginal but non-random and another (*C*) is completely unstable. Only two samples at site *B* reached clearly defined stable end-points[11]; results from that site will not be used in subsequent calculations. Samples from site *C*, a leucocratic granodiorite, changed direction erratically on repeat measurement, both before and after magnetic cleaning. Magnetic cleaning of test specimens from site *C* was pursued up to 800 oersted but no evidence of stable magnetic directions was found. The great variation in magnetic stability found in this study among rocks of similar origin and appearance is puzzling, but appears to be a common characteristic of granitic rocks[12].

Deutsch[13] calculated a mean Cretaceous pole for North America (68° N, 172.3° W) based on six independent palaeomagnetic investigations in areas ranging from New England and Quebec to California. The six poles are tightly grouped, with a precision parameter (κ) of 303 (ref. 10) and a circle of 95% confidence (α_{95}) with a radius of 3.9°. Substitution of later results for the Monteregian Hills intrusives of Quebec[14] and the addition of poles for Upper Cretaceous sedimentary rocks from Wyoming and Utah[15] and for alkalic igneous rocks from Arkansas[16] gives a mean North American Cretaceous pole of 69° N, 175° W, with precision parameter 574 and confidence interval 2.3°. This tight grouping is strong evidence for the existence of a dipolar magnetic field during Cretaceous time and further suggests that the Cretaceous was a time of little or no polar wandering or drift of the North American continent[17].

Fig. 2 illustrates the fact that poles for stable sites in Cretaceous granitic rocks from central Washington lie far outside the cluster of poles for the remainder of North America. This difference does not seem to be the result of incomplete averaging of secular variation. The large distances between sampling sites and the plutonic nature of the rock suggest a long magnetization time (radiometric ages span more than 10^7 yr but with large error limits). Most of the secular variation should thus be averaged out, even with a mean of only three sites. Moreover secular variation in the Cretaceous seems to have been less than at present[18] and judging from the analysis of the present field given by Cox[19,20] the probability of encountering current polar angular divergences of the observed magnitude (30 to 40°) due to secular variation alone is small. More likely explanations involve undetected tectonic effects, non-axial behaviour of the Cretaceous geomagnetic field or motion of the sampling area relative to the rest of North America.

The Cretaceous palaeolatitude of the Stevens Pass area predicted by its present location relative to the mean Cretaceous pole for North America is 56.5° whereas its palaeolatitude based on our palaeomagnetic measurements is nearer 30°. This difference in palaeolatitude corresponds to a difference in inclination of remanent magnetization of about 20 to 25°. Undetected tectonic tilting of Stevens Pass granitic rocks by some 25° upward to the north could thus account for the apparent palaeolatitude anomaly. Deformation of this magnitude, as by pervasive block faulting, would presumably be easy to detect but has apparently never been reported. The difference in palaeolatitude might also theoretically be the result of a period of anomalous behaviour of the geomagnetic field, such as the short period of geomagnetic fluctuations tentatively envoked by Spall[21] to account for anomalous palaeomagnetic directions in Mesozoic dolerites from Spitsbergen. Until the origin of the geomagnetic field is much better understood it will be impossible to rule out instances of such presumed breakdown of the dipole hypothesis; such breakdowns must, however, be rare and are particularly unlikely for the Cretaceous in view of its strongly dipolar magnetic field.

Difficulties with alternative interpretations tentatively lead us to favour an explanation based on plate tectonics. Thus if the Stevens Pass area and contiguous parts of the western Cordillera were originally part of a separate plate and were therefore free to move relative to North America until post-Cretaceous time, the observed palaeolatitude difference might easily be a consequence. In this model relative motion of the two continental masses ceased and welding of the two to form one continental land mass occurred at some time in the Tertiary. Before the welding of the plate, the interaction might have been of the convergent (subduction zone) or passive (transform) type. Such models are compatible with the analysis of faunal distributions given by Monger and Ross[1], and can also be fitted within the framework of current plate tectonic theories[3,4,22]. Fig. 2 shows the locus of possible poles of rotation of the Stevens Pass area relative to the rest of North America, based on our sparse palaeomagnetic data. This locus passes close to Morgan's pole of rotation of the Pacific plate[22] relative to the North American plate, but this may be fortuitous because, according to Atwater[3], the Kula and Farallon plates should have prevented any interaction between the Pacific and North American plates at the latitude of Seattle. Perhaps the Stevens Pass area once moved as part of the Kula plate[3,4]. At the estimated velocity of the Kula plate relative to North America (12 cm yr^{-1} in a generally northward direction), the Stevens Pass area could have increased in latitude by 25° in as little as 25 m.y.

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Reaction of Silica Gel with a Volatile Boron Compound released from Borosilicate Glass

DURING a study of the sorption of *n*-butanol and methanol vapours on silica gels in a vacuum system the formation of a by-product, identified as a silica–boron complex, was observed when the reaction was carried out in 'Pyrex' glassware. This by-product was not formed in a silica vessel and it was therefore concluded that the 'Pyrex' glassware was the source of the boron. Vacuum systems and techniques similar to those employed in this investigation are widely used in studying adsorption on oxide surfaces. It is thus important that workers in this field should be aware of the possible contamination of substrates by a volatile boron compound, and the consequent modification of the surface properties of the oxides.

The vacuum system comprised two interconnected flasks (each 50 cm^3), one for the sample and the other for the alcohol. The flasks had been washed thoroughly with hot distilled water. The gel samples were prepared as films by spreading 7 mg of powder, well ground by hand and heated to 500° C in air, over an area of 1 cm^2 and pressing at 1 t cm^{-2}. The films were contained in an aluminium holder with a 2 mm × 10 mm aperture and were placed in the sample flask in such a way that direct contact of the gel with the glass surface was avoided. The flask was heated to 140° C, evacuated to 10^{-4} torr and then exposed to the vapour from the liquid. The gel was exposed to the vapour overnight at 140° C and the physically adsorbed vapours were removed by evacuating again to 10^{-4} torr.

When films exposed to methanol vapour were analysed spectrographically (arc emission) using boron-free graphite electrodes they were found to contain up to 5,000 p.p.m. boron in the part of the material directly exposed to the vapour. Untreated films showed no detectable boron. The infrared spectrum of the films after exposure to alcohol or water vapour showed a strong band near 7.2 μm (Fig. 1) whose intensity increased in the order water, butanol, methanol (the order of increasing solubility of B_2O_3 in these solvents). The spectra of films treated in a silica reaction vessel did not show the 7.2 μm band. This observation, and the fact that the amount of boron released from the heated sample flask decreased with

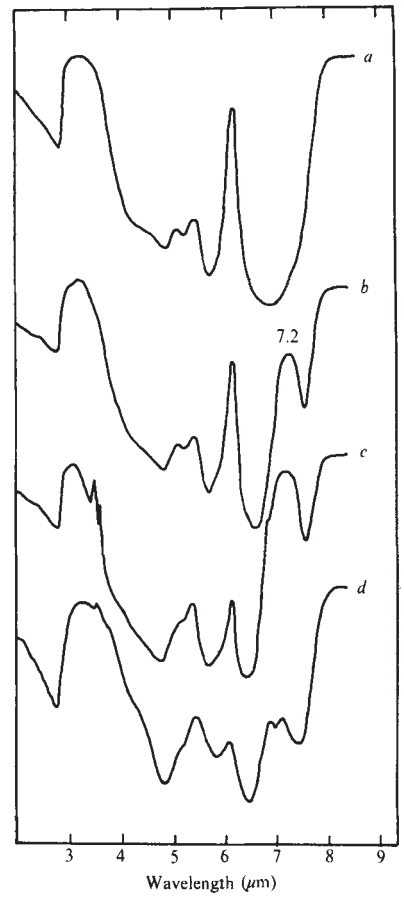
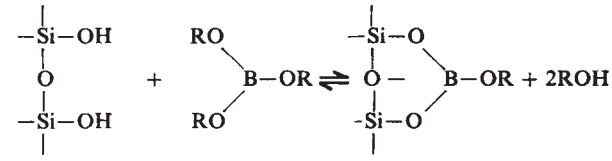


Fig. 1 Infrared spectrum of silica–boron complex. *a*, Fresh silica gel (pressed to a film); *b*, after water vapour treatment; *c*, after methanol vapour treatment; *d*, flushing with dry ammonia gas following methanol vapour treatment.

usage, shows that only the heated 'Pyrex' surface released boron. It is known that the volatility of oxides such as B_2O_3 is increased by the presence of water vapour at elevated temperatures and the species in the vapour phase is probably either metaboric acid BO(OH) or boric acid $B(OH)_3$. In the presence of alcohols, the acid will be esterified forming BO(OR) or $B(OR)_3$.

The 7.2 μm band can be ascribed to a surface group containing boron in three-fold coordination[2], probably resulting from condensation with silanol groups



Where R = H, methyl or n-butyl.

The silica–boron complex was unstable in water vapour, being partially hydrolysed at 80% RH and more completely in the presence of HCl gas or liquid water. The extent to which the complex forms, as shown by the intensity of the 7.2 μm band, increases with the specific surface area of the silica gel used (Table 1). The small contents of Na and K in the gels

Table 1 Relative Intensities (RI) of the 7.2 μm Band of Silica Gels with Differing Specific Surface Area

Preparation	Specific surface area ($m^2\ g^{-1}$)	RI	Na (%)	K (%)
Hydrolysis of tetraethyl silicate[3]	663	5	0.006	0.010
Precipitation of Na-silicate	423	4	0.024	0.006
'Aerosil' flame hydrolysis of $SiCl_4$	200	2.9	0.007	0.10

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